

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

Technical Presentation – Intelligent Agents in Real-World AI Systems

1. Reactive Agent-Based Traffic Control System

Slide 1: Title

Reactive Agent-Based Traffic Control System

Technology: AI Agents + IoT Sensors + Streaming Data + Smart Traffic Signals

Slide 2: Objective

- Monitor traffic conditions in real time.
 - Detect congestion and traffic density.
 - Automatically control traffic signals.
 - Reduce waiting time and traffic congestion.
 - Respond immediately without requiring long-term planning.
-

Slide 3: Reactive Agent

A Reactive Agent responds directly to the current state of the environment.

Input → Perception → Rule → Action

Example:

Traffic Sensors



Vehicle Density



Reactive Agent



Traffic Decision



Signal Change

Slide 4: System Architecture

Cameras / IoT Sensors



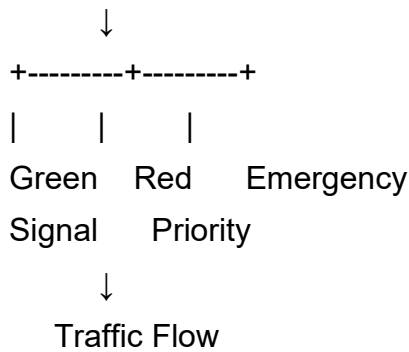
Streaming Data



Data Processing



Reactive Traffic Agent



Slide 5: Agent Decision Rules

Traffic Condition	Agent Action
Low traffic	Normal signal cycle
High traffic	Extend green time
Very high traffic	Give longer green phase
Empty road	Reduce green time
Emergency vehicle	Give priority

Example:

if traffic_density > 80:

 green_time = 60

elif traffic_density > 50:

 green_time = 45

else:

 green_time = 30

Slide 6: Streaming Data Processing

Traffic data continuously arrives from:

- CCTV cameras
- Vehicle sensors
- GPS devices
- IoT sensors
- Emergency vehicles

A streaming framework can process the data continuously.

Sensors → Stream → Processing → Agent → Signal



Slide 7: Congestion Minimization

The agent monitors:

- Vehicle count
- Queue length
- Average waiting time
- Traffic speed

The signal timing is continuously adjusted according to current traffic conditions.

Slide 8: Advantages

- Fast response.
- Real-time decision-making.
- Reduced congestion.
- Reduced waiting time.
- Supports emergency vehicles.

Slide 9: Conclusion

A Reactive Agent is suitable for traffic systems because it responds immediately to changing traffic conditions without requiring complex long-term planning.

2. Goal-Based Agent for Healthcare Diagnosis

Slide 1: Title

Goal-Based Agent for Healthcare Diagnosis

Technology: AI Agent + Clinical Data + Machine Learning + Decision Support

Slide 2: Objective

The goal-based agent aims to:

- Identify possible diseases.
 - Analyze patient information.
 - Achieve a diagnostic goal.
 - Select appropriate diagnostic pathways.
 - Support doctors in treatment decisions.
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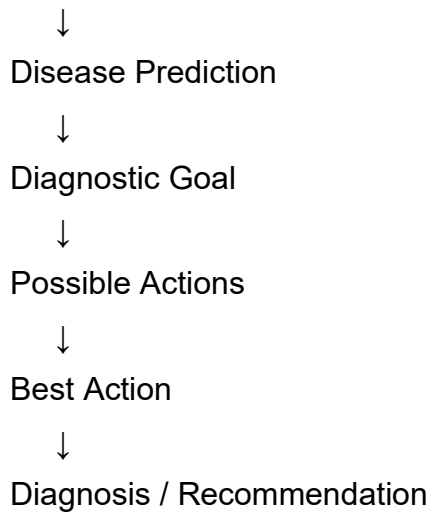
Slide 3: Goal-Based Agent

A Goal-Based Agent selects actions that help achieve a predefined goal.

Patient Data

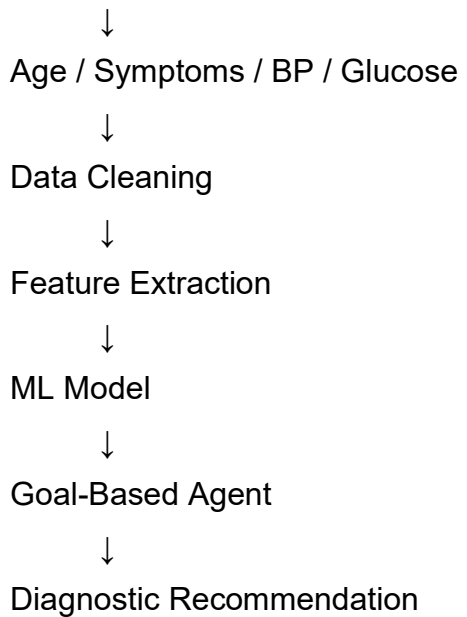


Agent Perception



Slide 4: Patient Data Pipeline

Electronic Health Records



Slide 5: Diagnostic Goals

Example goals:

- Detect diabetes.
 - Detect cardiovascular risk.
 - Identify abnormal glucose levels.
 - Recommend additional tests.
 - Reduce diagnostic uncertainty.
-

Slide 6: Decision-Making

Example:

Glucose High?

|

Yes

↓

Check BMI

|

BMI High?

|

Yes

↓

Check Family History

|

Yes

↓

High Diabetes Risk

The agent evaluates possible pathways and selects the one most likely to achieve the diagnostic goal.

Slide 7: Treatment Pathway

Patient

↓

Risk Assessment

↓

Low Risk —————→ Monitoring

↓

Medium Risk ———→ Lifestyle Intervention

↓

High Risk —————→ Doctor Review + Further Testing

The agent should act as a decision-support system, not an autonomous replacement for clinicians.

Slide 8: Advantages

- Faster analysis of patient information.
- Supports clinical decision-making.
- Can process large datasets.
- Provides consistent decision logic.
- Supports early detection.

Slide 9: Ethical Considerations

- Patient privacy.
- Data security.
- Bias prevention.
- Explainability.
- Human medical supervision.

Slide 10: Conclusion

A Goal-Based Agent is useful in healthcare because it evaluates patient information according to a specific diagnostic objective and selects actions that move the system toward that objective.

3. Utility-Based Agent for E-Commerce Recommendations

Slide 1: Title

Utility-Based Agent for E-Commerce Recommendations

Slide 2: Objective

- Understand customer preferences.
- Evaluate multiple products.
- Assign utility scores.
- Recommend the best products.
- Maximize customer satisfaction.

Slide 3: Utility-Based Agent

A utility-based agent evaluates different actions and selects the action with the highest expected utility.

User Profile



Candidate Products



Utility Calculation



Ranking



Highest Utility



Recommendation

Slide 4: Utility Factors

The agent can consider:

- Price
- Rating
- Brand preference
- Product relevance
- Previous purchases
- Discount
- Delivery time

Example:

$$[\\ U=0.30R+0.20P+0.20B+0.15D+0.15Q \\]$$

where:

- (R) = relevance
- (P) = price suitability
- (B) = brand preference
- (D) = discount
- (Q) = quality/rating

Slide 5: Example

Suppose three products have:

Product	Utility
---------	---------

Product A	0.72
-----------	------

Product B	0.91
-----------	------

Product C	0.65
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The agent recommends Product B because it has the highest utility.

Slide 6: Large-Scale Data Processing

User Clicks

Purchases

Searches

Ratings



Distributed Data Processing



User Profile



Utility Model



Product Ranking



Recommendations

PySpark can process millions of user-item interactions.

Slide 7: Agent Feedback

Recommendation



User Click



Purchase / Ignore



Feedback



Update Utility



Improved Recommendation

Slide 8: Advantages

- Highly personalized.
- Handles multiple preferences.
- Optimizes recommendations.
- Scales to millions of products/users.
- Continuously adapts to behavior.

Slide 9: Conclusion

A Utility-Based Agent is effective in e-commerce because it evaluates multiple factors simultaneously and selects recommendations that maximize expected user satisfaction.

4. Learning Agent for Personalized Education

Slide 1: Title

Learning Agent for Personalized Education System

Technology: AI + Learning Agent + PySpark + Adaptive Learning

Slide 2: Objective

- Monitor student performance.
 - Identify weak areas.
 - Adapt content difficulty.
 - Learn from student feedback.
 - Provide personalized learning paths.
-

Slide 3: Learning Agent Architecture

Student Activity



Data Collection



PySpark Processing



Performance Analysis



Learning Agent



Content Recommendation



Student Feedback



Agent Learning



Updated Recommendation

Slide 4: Student Data

The platform collects:

- Quiz marks.
- Assignment scores.
- Attendance.
- Time spent learning.

- Number of attempts.
- Video completion.
- Topic-wise performance.

Slide 5: Adaptive Difficulty

Performance > 80%



Advanced Content

Performance 50–80%



Normal Content

Performance < 50%



Remedial Content

The agent continuously changes the difficulty based on student performance.

Slide 6: Learning Process

A learning agent contains four important components:

1. Performance element – chooses learning actions.
2. Learning element – improves the model.
3. Critic – evaluates student feedback.
4. Problem generator – creates new learning opportunities.

Slide 7: PySpark Integration

Thousands of Students



Student Activity RDDs



Distributed Processing



Performance Statistics



Learning Agent



Personalized Learning

PySpark allows large numbers of student records to be processed efficiently.

Slide 8: Example

If a student repeatedly performs poorly in Python loops:

Poor Performance



Agent Detects Weakness



Basic Loop Lesson



Practice Questions



New Assessment



Performance Improved



Intermediate Content

Slide 9: Benefits

- Personalized education.
 - Real-time feedback.
 - Adaptive difficulty.
 - Early identification of weak topics.
 - Scalable to thousands of learners.
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Slide 10: Conclusion

A Learning Agent continuously learns from student interactions and improves its recommendations, making it highly suitable for adaptive and personalized education.

5. Multi-Agent System for Disaster Management

Slide 1: Title

Multi-Agent System for Disaster Management

Slide 2: Objective

- Detect disasters rapidly.

- Combine information from multiple sources.
- Coordinate emergency response.
- Generate early warnings.
- Support rescue operations.

Slide 3: Multi-Agent System

A Multi-Agent System consists of several autonomous agents that cooperate to solve a complex problem.

Sensor Agent

|

Satellite Agent ----+

|

|

Social Media Agent -+→ Decision Agent

|

↓

Communication Agent

|

Emergency Services

Slide 4: Types of Agents

Sensor Agent

Collects:

- Temperature
- Water level
- Seismic activity
- Wind speed

Satellite Agent

Analyzes satellite imagery for:

- Flood areas
- Landslides
- Forest fires

Social Media Agent

Detects disaster-related reports.

Decision Agent

Combines information and determines disaster severity.

Communication Agent

Sends alerts to authorities and the public.

Slide 5: Data Sharing

Sensor Data



Sensor Agent



+-----+



Satellite → Decision Agent ← Social Media



Data Fusion



Disaster Level

Data fusion improves confidence by combining independent evidence.

Slide 6: Decision-Making

Example:

Water Level ↑

+

Heavy Rainfall

+

Satellite Flood Detection



High Flood Probability



Decision Agent



Emergency Warning

Slide 7: Coordination Strategy

Agents communicate through:

- Shared data repositories.
- Message passing.

- Event notifications.
- Priority-based communication.

Example:

Sensor Agent → Decision Agent

Decision Agent → Communication Agent

Communication Agent → Emergency Services

Emergency Services → Response Agents

Slide 8: Real-Time Response

Disaster Detected



Severity Estimated



Area Identified



Emergency Alert



Evacuation Recommendation



Rescue Coordination

Slide 9: Advantages

- Parallel monitoring.
 - Faster disaster detection.
 - Distributed decision-making.
 - Multiple data sources.
 - Improved emergency coordination.
 - Scalable architecture.
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Slide 10: Overall Conclusion

Multi-Agent Systems are particularly suitable for disaster management because no single agent needs to perform every task. Specialized agents can independently monitor sensors, analyze satellite information, process public reports, make decisions, and coordinate emergency responses.

Final Comparison

Agent Type	Application	Main Decision Mechanism	Key Benefit
Reactive Agent	Traffic control	Condition → Immediate action	Fast response
Goal-Based Agent	Healthcare	Select action to achieve goal	Goal-oriented diagnosis
Utility-Based Agent	E-commerce	Maximum utility	Personalized optimization
Learning Agent	Education	Learn from feedback	Continuous adaptation
Multi-Agent System	Disaster management	Agent collaboration	Distributed coordination

Key AI Concept

Reactive Agent



Responds to current situation

Goal-Based Agent



Chooses action to achieve goal

Utility-Based Agent



Chooses best-valued action

Learning Agent



Improves through experience

Multi-Agent System



Multiple agents cooperate

These five designs demonstrate how different intelligent-agent architectures can be matched to different real-world problems, while technologies such as PySpark, IoT, streaming analytics, machine learning, and distributed processing provide the data-processing infrastructure required for large-scale AI applications.

